Artefact 2: Production Traveller OCR Proof of Concept

Data Analyst Internship - ST Engineering

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Confidentiality note: All traveller details, identifiers and results shown here are mock data. No company document, employee information, production record or system screen is included.

1. Description of the deliverable

This artefact documents a small proof of concept for extracting selected fields from a scanned production traveller. It includes the field mapping, the proposed OCR workflow, a mock structured output, and the result of comparing two ways to identify the operator.

My contribution: I studied the document structure, mapped the required fields, tested OCR outputs, compared the operator-identification methods, and documented the recommended approach.

How it will be used: The proof of concept can help assess whether repetitive data entry can be reduced while keeping manual review for uncertain or invalid values.

2. Mock traveller field mapping

Document field	Mock value	Output field	Validation
Job Order	DEMO-JO-001	job_order	Required
Part Number	DEMO-PART-A	part_number	Required
Serial Number	DEMO-001	lot_serial	Required
Operation	T010	operation	Check against operation list
Operation Name	Kit Inspection	operation_name	Text comparison
Work Centre	DEMO-WC	work_centre	Reference lookup
Completion Date	01 Jul 2026	completion_date	Valid date and range
Operator Stamp	1234	stamp_id	Numeric pattern and lookup

3. Proposed OCR workflow

- Capture: Receive the scanned traveller pages and separate them into individual documents.
- Extract: Read the header fields, operation details, dates and stamp area.
- Validate: Check the field formats and flag missing, overwritten or low-confidence values.
- Structure: Save only the reviewed values in the required output fields.

4. Mock structured output


```
{
  "job_order": "DEMO-JO-001",
  "part_number": "DEMO-PART-A",
  "lot_serial": "DEMO-001",
```

```
    "operation": "T010",
    "work_centre": "DEMO-WC",
    "completion_date": "2026-07-01",
    "stamp_id": "1234",
    "validation_status": "REVIEW_REQUIRED"
  }
```


5. OCR method comparison

Method tested	Observed result	Decision
Operator name from curved stamp	The characters were inconsistent and the name could not be reproduced reliably.	Not selected for the proof of concept.
Numeric stamp ID	The numeric identifier was recognised more consistently in the test.	Preferred method, subject to wider testing and an authorised lookup.

6. Technical skills and competency level

Technical skill / competency	Level	Evidence and area for improvement
Data Collection and Analysis	Level 2 - Apply	I mapped document values to structured fields and added validation rules.
Software Testing	Level 2 - Apply	I compared two OCR approaches and recorded the test outcome before deciding which method to use.
Process Improvement	Level 2 - Apply with guidance	I proposed a workflow that reduces repetitive entry while keeping review controls.

7. Generic skills used

Critical Core Skill	Task / action	Result
Problem Solving	Investigated why names from curved stamps were unreliable.	Found a simpler and more consistent identifier.
Digital Fluency	Used UiPath OCR and structured output formats.	Created a method that can be tested again.
Sense Making	Compared OCR output with the visible document context.	Avoided accepting text that looked possible but was incorrect.
Communication	Documented the limitation and the proposed lookup method.	Made the recommendation clear to both technical and operational users.

8. Corrective action and reflection

The operator-name OCR was inconsistent, so I tested the numeric stamp ID instead and obtained a more repeatable result. This taught me that a simpler method is more useful when it can be checked consistently. The next step is to test a larger approved sample against an authorised master list while keeping human review for exceptions.